***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 26   Innate Immunity: Broadly Specific Host Defenses

26.1   Multiple Choice Questions

1) The ability of humans to resist a disease is called
A) dormancy.
B) immunity.
C) resistance.
D) susceptibility.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

2) Cells that can engulf foreign particles, and can ingest, kill, and digest most bacterial pathogens are called
A) red blood cells.
B) phagocytes.
C) reticulocytes.
D) resistant cells.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

3) Adaptive immunity occurs when
A) death results from pathogen infection.
B) the innate immune response is overly effective.
C) a broad, rapid response is needed to a wide range of pathogens from the body regardless of the specific type of pathogen.
D) the innate immune response fails to eliminate pathogens in the body and virulent infections persist after the initial innate defense response.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

4) Blood and lymph have the following in common EXCEPT
A) nucleated cells.
B) proteins.
C) red blood cells.
D) none of these.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

5) Stem cells are produced and developed in the
A) bone marrow.
B) brain.
C) liver.
D) stems of plants.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

6) Which of the following is TRUE about cytokines?
A) They are carbohydrates.
B) They play an important role in immunity, but do not affect stem cell growth.
C) They play a role in the growth of stem cells, but are not important in immunity.
D) They are proteins that play an important role in immunity and the growth of stem cells.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

7) Which of the following cell types has NO nucleus?
A) erythrocytes
B) lymphocytes
C) phagocyte
D) All of these cells lack a nucleus.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

8) The secondary lymphoid organs consist of which of the following?
A) lymph nodes only
B) mucosal-associated lymphoid tissue (MALT) only
C) the spleen only
D) lymph nodes, MALT, and the spleen
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

9) Which of the following can cause allergy symptoms and inflammation by degranulation?
A) dendritic cells
B) erythrocytes
C) macrophages
D) polymorphonuclear leukocytes
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

10) The unique antigen-reactive proteins of T cells are
A) antibodies.
B) immunoglobins.
C) T cell receptors.
D) None of these are correct.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

11) The rapid increase in adaptive immunity after a second antigen exposure is called
A) immune memory.
B) specificity.
C) tolerance.
D) immune memory, specificity, or tolerance.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

12) An example of pathogen-associated molecular patterns (PAMP) is
A) cell-surface proteins.
B) flagellin proteins.
C) lipopolysaccharide (LPS).
D) pilus.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.2

13) The process by which antibodies block interactions between pathogens or their products and host cells is termed
A) attenuation.
B) complement.
C) interference.
D) neutralization.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

14) Which of the following are molecular mediators of inflammation?
A) immunoglobulins
B) lipopolysaccharide
C) erythrocytes
D) both chemokines and cytokines
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

15) Immunoglobulins are produced by B cells and are also known as
A) antibodies.
B) antigens.
C) bacteria.
D) pathogens.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.2

16) MHC I proteins are found
A) on B cells only.
B) on macrophages only.
C) on dendritic cells only.
D) on all nucleated cells.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

17) Enhanced phagocytosis of antibody-sensitized cells is known as
A) complementation.
B) immunization.
C) opsonization.
D) tolerance.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

18) Pattern recognition receptors are most directly used
A) by granulocytes to trigger degranulation.
B) by neutrophils to detect self tissues.
C) by phagocytes to detect pathogens.
D) by B cells to produce antibodies.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

19) Which of the treatments listed below would be most effective for a patient with a genetic defect that causes severe combined immunodeficiency (SCID), a condition in which there is a severe deficiency in B and T lymphocytes?
A) intravenous antibiotics
B) bone marrow transplant
C) multiple immunizations
D) repeated doses of antisera
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.4

20) The body′s non-inducible, preexisting ability to recognize and destroy a variety of pathogens or their products is called
A) adaptive immunity.
B) innate immunity.
C) phagocytosis.
D) cytotoxic response.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

21) Adaptive immune responses are directed at pathogen molecules called
A) antigens.
B) antibodies.
C) T-cell receptors.
D) PAMP.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

22) The cells active in both innate and adaptive immunity develop from common pluripotent precursors in the bone marrow called
A) B cells.
B) killer cells.
C) stem cells.
D) leukocytes.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

23) The first defense cells that interact with a pathogen in the body are
A) T cells.
B) macrophages.
C) B cells.
D) mast cells.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

24) The inflammatory cytokines IL-1, IL-6, and TNF-α are also
A) opsonizing agents that coat foreign cells.
B) responsible for decreasing vascular permeability.
C) capable of producing systemic fever by stimulating the release of prostaglandins in the brain.
D) part of the classical pathway of complement activation.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

25) DiGeorge′s syndrome is a developmental defect that prevents the maturation of the thymus. What cell type would be reduced by this condition?
A) T cells
B) B cells
C) macrophages
D) lymphocytes
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.4

26) Intracellular pathogens sometimes produce ________ that can kill phagocytes.
A) lipopolysaccharides
B) PAMPs
C) PRRs
D) leukocidins
Answer: D
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.7

27) C-reactive protein is a
A) pathogen-associated molecular pattern (PAMP).
B) homotrimeric adaptor protein.
C) lipoprotein.
D) pattern recognition receptor (PRR).
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

28) Pattern recognition receptors (PRRs) are found on which of the following cells?
A) macrophages
B) bacteria
C) neutrophils
D) macrophages and neutrophils
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

29) Which of the following interact with toll-like receptors (TLRs)?
A) bacterial LPS
B) immunoglobulins
C) major histocompatibility complex proteins
D) none of these
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

30) Interaction of a pathogen-associated molecular pattern (PAMP) with a pattern recognition receptor (PRR) results in
A) formation of transmembrane pores that cause cell lysis.
B) transmembrane signal transduction that initiates transcription of genes involved in phagocytosis, inflammation, and pathogen killing.
C) molecular activation of the adaptive immune system.
D) a superantigen reaction that can cause septic shock.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.6

31) Which of the following are proteins that interact directly with antigens during the adaptive immune response?
A) immunoglobins
B) major histocompatibility complex
C) T cell receptors (receptors on T cells)
D) all of these
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

32) Communication between cells of the immune system is accomplished in many cases through
A) allelic exclusion.
B) clonal deletion.
C) cytokines.
D) allelic exclusion, clonal deletion, and cytokines.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.5

33) Chemokines are a group of small proteins that
A) attract T cells to sites of injury.
B) function as chemoattractants for phagocytes and lymphocytes.
C) potentiate specific immune responses.
D) attract T cells, phagocytes, and lymphocytes, as well as potentiate specific immune responses.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.5

34) The PAMP recognized by mannose-binding lectin (MBL) is the sugar mannose, found as a repeating subunit in
A) human mucus.
B) lymphocyte receptors.
C) bacterial polysaccharides.
D) bacterial proteins.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

35) Which of the following is a likely target for a toll-like receptor (TLR)?
A) tumor necrosis factor
B) interleukin IV
C) flagellin
D) insulin
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

36) Interaction of a PAMP with a TLR triggers transmembrane signal transduction and subsequent
A) clonal expansion.
B) antibody production.
C) complement protein activation.
D) phagocytosis and inflammation.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

37) Signal transduction pathways initiate activation of transcription after specific
A) T_H cell death.
B) ligand-receptor binding on the cell surface.
C) cytokines diffuse across the cytoplasmic membrane.
D) antigen-antibody binding.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

38) Some of the cytokines produced by lymphocytes are called
A) interleukins.
B) pathogen recognition receptors.
C) complement.
D) epitopes.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

39) Fever is induced at the systemic level by ________, which is an endogenous pyrogen.
A) CXCL8
B) IL-12
C) IL-6
D) CCL2
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

40) The primary function of a phagocyte is to
A) destroy pathogens.
B) engulf pathogens.
C) evade pathogens.
D) both engulf and destroy pathogens.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

41) Which of the following are NOT phagocytes?
A) dendritic cells
B) monocytes
C) neutrophils
D) erythrocytes
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

42) Which of the following is a function of dendritic cells?
A) antigen presentation
B) plaque formation
C) phagocytosis
D) both phagocytosis and plaque formation
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

43) Phagocytes have a pathogen-recognition system known as ________ that leads to the recognition, containment, and destruction of a pathogen.
A) collagen
B) fibrin
C) pattern recognition receptors (PRR)
D) pathogen associated molecular pattern (PAMP)
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

44) Oxygen compounds toxic to pathogens include
A) hydrogen peroxide.
B) hypochlorus acid.
C) nitric oxide.
D) hydrogen peroxide, hypochlorus acid, and nitric oxide.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

45) The enhancement of phagocytosis due to deposition of antibody on the surface of a pathogen or antigen is called
A) complementation.
B) opsonization.
C) inflammation.
D) antibody class switching.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

46) ________ are cytotoxins produced by T cells that cause apoptosis.
A) Perforins
B) Granzymes
C) Phagocytes
D) Macrophages
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

47) Another name for "programmed cell death" is
A) necrosis.
B) perforation.
C) apoptosis.
D) cellular degradation.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

48) *Streptococcus pyogenes* produces proteins called ________, which alter(s) the surface of the pathogen and inhibits phagocytosis.
A) glycolipids
B) leukocidins
C) M proteins
D) pus
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

49) Some intracellular pathogens produce phagocyte-killing proteins called ________ that kill the phagocyte after ingestion of the pathogen.
A) antibodies
B) antigens
C) leukocidins
D) pus
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

50) The first cell type active in the innate response is usually a(n)
A) phagocyte.
B) erythrocyte.
C) fibroblast.
D) antibody.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

51) When dendritic cells ingest antigen, they migrate to the ________, where they present the antigen to T lymphocytes.
A) kidneys
B) lymph nodes
C) thymus
D) spleen
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

52) The increased rate of O_2 uptake by activated phagocytes is called the
A) hypoxemia.
B) pyogenesis.
C) respiratory burst.
D) peroxidase.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

53) Which of the following uses its cell wall glycolipids to absorb hydroxyl radicals and superoxide anions?
A) *Staphylococcus aureus*
B) *Streptococcus pyogenes*
C) *Mycobacterium tuberculosis*
D) *Escherichia coli*
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 26.7

54) Organisms such as *Streptococcus pyogenes* and *Staphylococcus aureus* that produce leukocidins and are associated with pus are called ________ pathogens.
A) pyogenic
B) radical
C) enterotoxigenic
D) hemorrhagic
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

55) A(n) ________ is a group of sequentially interacting proteins important in innate and adaptive immunity.
A) complement
B) opsonin
C) inflammation
D) regeneration
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

26.2 True/False Questions

1) Immunity results from the actions of cells that circulate throughout the body, primarily through the blood and lymph.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

2) Erythrocytes are the most numerous cells in human blood.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

3) Anticoagulants promote the clotting of blood.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

4) Blood serum contains cells and clotting proteins.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.3

5) Monocytes and granulocytes are the two lineages of myeloid cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

6) Antibodies are soluble proteins produced by T cells.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

7) All gram-negative bacteria have lipopolysaccharides in their outer membranes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

8) Dendritic cells are phagocytes with antigen-presenting properties.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

9) In adaptive immunity, pathogen specific receptors are produced in large numbers only after exposure to the pathogen or its products.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.1

10) The T cell, with its T cell receptor, can recognize antigens only when the antigens are complexed with self proteins known as major histocompatibility complex found on host cell surfaces.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

11) Phagocytes interact speedily and effectively with pathogens because they have evolved specialized molecules called pattern recognition receptors (PRRs) that interact directly with PAMPs.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

12) Mannose on mammalian cells is inaccessible to mannose-binding lectin.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

13) All TLR can react with only one specific PAMP.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

14) Antigen receptors can directly connect to signal transduction pathways because immunoglobulins and TCRs have very small cytoplasmic domains.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

15) The human leukocyte antigen spans about 4 Mbp on human chromosome 6.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

16) Antibodies are insoluble proteins.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

17) The cytoplasmic domains of immunoglobulins and TCRs have cytoplasmic tyrosines that can be phosphorylated.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

18) The presence of neutrophils in higher than normal numbers in the blood or at a site of inflammation indicates an active response to a current infection.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

19) Monocytes are circulating precursors of macrophages and dendritic cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.5

20) PRRs were FIRST observed in phagocytes in *Drosophila* (where they are called Toll receptors).
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

21) Inflammation is the usual outcome of an adaptive immune response but not an innate immune response.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.8

22) Some *Mycobacterium species* produce carotenoids that neutralize singlet oxygen and prevent killing.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

23) Dead phagocytes make up much of the material of pus.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

24) Localized infections by pyogenic bacteria often form boils or abscesses.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

25) Bacterial capsules enhance the adherence of phagocytes to bacterial cell walls, thereby promoting phagocytosis.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.7

26) Nonencapsulated strains of *Streptococcus pneumoniae* are highly virulent.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.7

26.3 Essay Questions

1) How does the body respond when it first encounters a foreign particle or bacterial pathogen?
Answer: The body has defensive cells called phagocytes (neutrophils and macrophages are examples) that recognize foreign particles and bacterial pathogens. Initial interactions with pathogens or foreign particles recruit large numbers of phagocytes to the site of infection. Here the phagocytes activate defense genes, leading to the transcription and translation of proteins that destroy the pathogen or foreign particle. Macrophages also present the antigens to T helper cells, which starts the process of activating the adaptive immune system.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.4

2) Compare and contrast the origin and roles of B cells and T cells.
Answer: Both B cells and T cells are called lymphocytes. They both originate in the bone marrow. However, B cells mature in the bone marrow while T cells mature in the thymus. B cells are precursors of antibody-producing plasma cells; T cells are not. T cells directly interact with and can lysis infected human cells, while B cells produce antibodies and do not directly kill infected cells.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.4

3) How do phagocytes interact speedily and effectively with pathogens?
Answer: Phagocytes have evolved specialized molecules called pattern recognition receptors (PRR) that interact directly with specific pathogen-associated molecular patterns (PAMPs) to activate these pathogens to ingest and destroy the targeted pathogens by phagocytosis.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

4) Explain why an individual lymphocyte clone that interacts with a specific LPS constituent of *Salmonella* will not react with the LPS of other bacteria.
Answer: The adaptive response is inducible only when triggered by a unique antigen on a pathogen. The terminal sugars that constitute an antigen on the polysaccharides of *Salmonella spp*. are unique for the genus and are not shared by other bacteria.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

5) What is antigen presentation? Why is antigen presentation important for long-term immunity?
Answer: Antigen presentation is the transportation and display of MHC-embedded peptides to the phagocyte surface after ingestion and degradation to small peptides by APCs. The MHC-peptide complex is then the target for T helper cells that will activate other immune cells in the adaptive immune response. This immune response will be specifically targeted towards the antigen that was presented by the phagocyte. The adaptive immune response generates memory B cells and T cells that provide long-lasting immunity towards any foreign particle that has this antigen.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

6) Why do individuals with genetic defects that prevent neutrophil or macrophage development usually die at an early age? Be specific and detailed in your answer to demonstrate your knowledge on the role of these cells in the immune system.
Answer: Individuals with such defects cannot live without extraordinary intervention, such as isolation from all environmental exposure. Neutrophils and macrophages are phagocyctic and antigen presenting cells. Individuals without these cells due to genetic defects or bone marrow suppression have a weakened innate immune response that can allow normal environmental bacteria and fungi to cause infections. The adaptive immune response of these individuals is also abnormal because neutrophils and macrophages are required for antigen presentation, which is the first step in the adaptive immune response that begins the selection and stimulation of clonal lymphocytes. Without the activation of the adaptive immune response, no lasting immunity to any pathogen will develop and, thus, individuals with genetic defects in neutrophil or macrophage maturation are severely immune-deficient. In a normal environment, they develop recurrent infections from bacteria, viruses, and fungi, which results in their death.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.4

7) Describe the signal transduction pathway that is activated when LPS binds to TLR-4.
Answer: After LPS binds to TLR-4, TLR-4 then binds proteins in the cytosol, starting a cascade of reactions that activates transcription factors such as nuclear factor kappa B (NFKB), initiating transcription of downstream genes. Many of the NFKB-regulated genes encode host response proteins, such as the cytokines that activate cells and initiate inflammation.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

8) Predict the consequence for an individual of a deleterious genetic mutation in the gene for TLR-4.
Answer: A mutation that changes TLR-4 would hamper the immune system′s ability to recognize and respond to gram-negative bacteria that contain LPS. This could result in an increased risk for infection by gram-negative bacteria.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 26.6

9) Briefly describe how a phagocyte engulfs and ingests a pathogen.
Answer: Phagocytes trap pathogens on surfaces such as blood vessel walls or fibrin clots. The membrane surrounding the pathogen pinches off and forms a phagosome, which moves inside the phagocyte and fuses with a lysosome to form a phagolysosome. The toxic substances and enzymes inside the phagolysosome usually kill and digest the engulfed microbial cell.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.6

10) How is immune memory beneficial to a host organism?
Answer: Immune memory provides the host with the ability to immediately resist previously encountered pathogens. After the initial encounter with an antigen, the host is stimulated to produce effective antigen-activated antibodies, or TCRs. A subsequent exposure to the same antigen stimulates rapid production of large quantities of the same T cells or antibodies for future protection.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 26.1

11) Briefly describe the effect of deposition of antibody or complement on the surface of a pathogen or antigen in relation to opsonization.
Answer: Opsonization is the enhancement of phagocytosis due to the deposition of antibody or complex on the surface of a pathogen or other antigen. If complement binds to an antibody-antigen complex on the cell surface, the cell is even more likely to be ingested. This is because most phagocytes, including neutrophils, macrophages, and B cells, have antibody receptors (FcR) as well as C3 receptors (C3R). These receptors bind the antibody constant domain and C3 complement protein respectively. Normal phagocytic processes are enhanced about tenfold by antibody binding and amplified another tenfold by C3 binding.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.10

12) Briefly describe classical complement activation and cell damage.
Answer: Classical activation of complement occurs when IgG or IgM antibodies bind (fix) antigens, especially on cells surfaces. The complement proteins act in a defined sequence with activation of one complement component leading to the activation of the next, and so on. Cl components (Clq, Clr, and Cls) then bind to the antibody-antigen complex, leading to C4-C2 deposition on at an adjacent membrane site. This complex is a C3 convertase, an enzyme that cleaves C3 to C3a and C3b. The C3b cleavage product then binds to the convertase, forming a complex that initiates a C5-C6-C7 interaction at a second membrane site. C8 and C9 are then deposited with the C5-C6-C7 complex, resulting in membrane damage and cell lysis.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 26.9

13) How do NK cells discriminate between normal, healthy cells and virus-infected or tumor cells?
Answer: NK cells use special MHCI receptors to recognize MHCI proteins on normal, healthy cells. Binding of the NK receptors to MHCI deactivates the NK cell, turning off the perforin and granzyme killing mechanisms. Many tumor and virus-infected cells reduce or eliminate MHCI protein expression patterns to evade the antigen-specific immune response. In the absence of deactivation of NK cells by MHCI proteins, the stress receptors on NK cells engage stress proteins on target cells releasing perforins and granzyme to destroy these target cells.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 26.10